

Applying AI to Systematic FX Trading: Validation and Deployment of a Counter-Trend Zone Strategy

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Abstract

This report documents the design, statistical validation, and live deployment of an automated foreign-exchange trading agent built around a single counter-trend strategy: fading H4 supply/demand zone touches *against* the prevailing daily trend (`zone_d1_against`). Artificial intelligence methods were applied not as a black-box predictor of price, but as a rigorous research pipeline—hypothesis generation, ablation screening, holdout testing, walk-forward validation, and cross-pair replication—to separate genuine edge from overfitting. After six of seven initial ICT-style concepts failed multiplicity-corrected significance tests, one cell survived a full validation chain and was deployed on a three-pair demo portfolio with strict risk constitution. Early live results (June 2026) are inconclusive but consistent with backtested expectancy. The system is observation-heavy: near-miss vaults, loss journals, and extension-target ladders collect data for future studies without altering validated trade logic.

1 Introduction

Retail and systematic traders frequently deploy complex multi-indicator systems without pre-registered validation. The present work inverts that pattern: **one strategy, one locked parameter set, one validation chain**. The research question is operational, not philosophical:

Does fading H4 zone touches against the D1 trend produce positive expectancy after transaction costs, across time and currency pairs, under walk-forward and holdout discipline?

The agent implements this hypothesis mechanically. AI-assisted engineering built the back-test harness, ablation grid, and live runner; human judgment locked parameters and governs any future promotion.

2 Related Work and Design Rationale

The strategy belongs to the ICT (Inner Circle Trader) family of supply/demand and higher-timeframe alignment concepts. Rather than deploying the full ICT vocabulary, we tested seven concepts individually (BOS, CHoCH, FVG, liquidity sweeps, order blocks, premium/discount, zone alignment) and retained only what survived statistical gates. Counter-trend zone fading against D1 bias emerged as the sole robust cell—with-trend variants of the same logic failed, confirming that the edge is mean-reversion specific, not generic “zone trading.”

3 Methodology

3.1 Strategy definition

Locked parameters (frozen before out-of-sample tests):

- Higher-timeframe alignment: D1, mode **against**
- HTF lookback: 10 bars; minimum D1 move: 60 pips
- Entry: H4 zone touch; fixed take-profit at 1.5R (never grid-swept)
- Soft stop at strategy level; broker catastrophe stop at $2.5\times$ soft

3.2 Validation pipeline

1. **Concept ablation** (7 ICT concepts): bootstrap p -values, Benjamini–Hochberg FDR at 5%; six concepts rejected.
2. **Holdout split**: in-sample 2015–2022, out-of-sample 2023–2025.
3. **Walk-forward**: seven windows, 4-year IS / 1-year OOS; EURUSD H4/all: 7/7 positive OOS windows, median +11.34 pips/trade.
4. **Cross-pair replication** (frozen parameters, scaled costs): GBPUSD +10.24 pips/trade ($p = 0.001$, 11/11 years); USDCAD +4.63 pips/trade ($p = 0.028$, 10/11 years); AUDUSD and NZDUSD excluded.
5. **Sealed look** (2026 YTD): 16 trades, +7.75 pips/trade, $p = 0.29$ — inconclusive, monitoring continues.

3.3 Live architecture

One process per symbol; deployment router maps (symbol, timeframe, session) $\rightarrow$ risk scale. Deployed cells: EURUSD H4 @ 1.0, GBPUSD H4 @ 0.5, USDCAD H4 @ 0.5. Adaptive position sizing uses conviction band 0.5–2% of balance $\times$ route scale. Unknown cells fail-safe to skip.

4 Results

Table 1: Validation summary (EURUSD anchor pair)

Metric	Value	Notes
Walk-forward OOS windows positive	7/7	H4/all cell
Median OOS pips/trade	+11.34	EURUSD
GBPUSD replication	+10.24	$p = 0.001$
USDCAD replication	+4.63	$p = 0.028$
Sealed 2026 (partial)	+7.75	$p = 0.29$, $n = 16$

4.1 Live deployment (June 2026)

Demo account (\$1000, Exness MT5) runs three symbol tabs. First confirmed manual demo fill: EURUSD long, TP hit (+28.10 USD). Automated orders require MT5 “Algo Trading” enabled; retcode 10027 rejections are logged when disabled.

5 Limitations

- `target_rr=1.5` was never grid-swept; all validation is conditional on this default.
- Sealed-period sample is small ($n = 16$); no claim of live superiority yet.
- Single strategy family; portfolio diversification is across pairs, not alphas.
- Extension-target ladder and near-miss vaults are hypothesis-generating only—they do not alter live orders.

6 Conclusion

Applying AI to FX trading here meant building a *validation apparatus*, not a neural network that “learns” from logs. One counter-trend zone cell survived a pre-registered gauntlet and is deployed under strict risk limits. The honest outcome is positive but modest: robust in backtest and walk-forward, promising in early demo, not yet proven live at scale. Future work (parked until trigger conditions): `target_rr` study, partial take-profit policies, USD-exposure manager, and autonomy-ladder research runners—all gated by human approval and the same validation pipeline.

References

- Project documentation: `docs/00-journey.md`, `docs/CHECKPOINT.md`
- Evidence artifacts: `docs/reviews/` (immutable snapshots)
- Live runbook: `docs/runbooks/vmware-windows.md`